Lecture 10

Machine Learning 1

Intelligent Systems
Vrije Universiteit

Overview

1 Classification

1.1 Naive Bayes Classifier

1.2 *k*-Nearest Neighbour

2 Introduction to Machine Learning

2.1 Types of ML Tasks

Part 1: Classification

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Classification

Classification is a machine learning problem that requires assigning a class label to a data point.

e.g. Can we predict whether a student will pass a course or not?

training dataset

	WorkGroup	Lecture	Pass
John	F	F	F
Peter	F	F	F
Marie	F	T	F
Stefan	F	T	T
Elisa	T	F	F
Mohammed	T	F	T
Rachel	T	T	F
…	T	T	T
	feature	feature	class label

Variants of Classification Problems

Features

Usually numerical or categorical

Binary classification

2 classes, *for example* Pass and Enjoy Course

Multiclass classification

>2 classes, *for example* Pass and Enjoy Course and Grade

Multilabel classification

0, some, or all classes per example, *for example* class color with 7 options

Classification

Prediction: what is the probability that Leo passed IS if he did not attend workgroups or lectures?

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

$$P(\text{ pass } \mid \neg wg, \neg l) = \frac{P(\text{ pass }, \neg wg, \neg l)}{P(\neg wg, \neg l)} = \frac{0.05}{0.25} = 0.2$$

$$P(\neg \text{pass } \mid \neg wg, \neg l) = \frac{P(\neg \text{pass }, \neg wg, \neg l)}{P(\neg wg, \neg l)} = \frac{0.2}{0.25} = 0.8$$

Note: the denominator does not depend on pass or ¬pass.

Probabilistic Classifier

We can use a probabilistic classifier to solve these types of problems. Main idea:

Input:

- Set of features (height, age)
- Set of classes (male, female)
- Examples (feature values, class) (age=20, height=183, male)

Output:

- Class *C* with the highest probability given the features for a *new* example

Joint Probability Table *Revisited*

How to calculate

P(pass|wg,l,¬p,¬b,m)?

WG	L	Pass	Understand	Bored	Gender	Pass
F	F	F	F	F	F	T
F	F	F	F	T	F	T
F	F	F	T	F	T	F
F	F	F	T	T	T	F
F	F	T	F	F	F	T
F	F	T	F	T	F	F
F	F	T	T	F	F	T
F	F	T	T	T	T	T
F	T	F	F	F	F	F
…						

Part 1.1: Naive Bayes Classifier

Towards a Bayesian Classifier *1*

We can express the probability of a certain class *C* given that we observe features *F*

$$p(C \mid F_1, \ldots, F_n)$$

Using Bayes Theorem

$$\boxed{P(A \mid B) = \frac{P(B \mid A) * P(A)}{P(B)}}$$

$$p(C \mid F_1, \ldots, F_n) = \frac{p(C)\, p(F_1, \ldots, F_n \mid C)}{p(F_1, \ldots, F_n)}$$

Towards a Bayesian Classifier 2

$$p(C|F_1, \ldots, F_n) = \frac{p(C)p(F_1, \ldots, F_n|C)}{p(F_1, \ldots, F_n)}$$

Rewrite the numerator using the chaining rule and we get

$$\frac{p(C)p(F_1|C) * p(F_2|C, F_1) * p(F_3|C, F_1, F_2) * \ldots * p(F_n|C, F_1, \ldots, F_n)}{p(F_1, \ldots, F_n)}$$

Unfortunately, this is very expensive to calculate.

Towards a Bayesian Classifier *3*

To decrease cost, Naive Bayes assumes conditional independence of all features

i.e. F_i is conditionally independent from F_j

$$p(F_i \mid C, F_j) = p(F_i \mid C)$$

Resulting in new formula:

$$p(C, F_1, \ldots, F_n) = p(C) * p(F_1 \mid C) * p(F_2 \mid C) \ldots$$
$$= p(C) \prod p(F_i \mid C)$$

Assumption of conditional independence is why the classifier is called naive.

Naive Bayes Classifier

Pick the most likely class

Conditionals

$$\text{classify}\,(f_1, \ldots, f_n) = \text{argmax}_c\, p(C = c) \prod_{i=1}^{n} p\,(F_i = f_i \mid C = c)$$

Document with features $f_1,\ldots,f_n$

Prior probability of class *C*

What happened to the denominator?

Naive Bayes Classifier

The Naive Bayes classifier may perform well even in cases when conditional independence assumptions are incorrect.

Classification problems require predicting the right class label, *not* accurately estimating probabilities.

Correct estimation $\Rightarrow$ accurate prediction

Accurate prediction $\not\Rightarrow$ correct estimation

Naive Bayes *Example*

Let's build a classifier than decides whether a sentence belongs to the course Intelligent Systems or Programming.

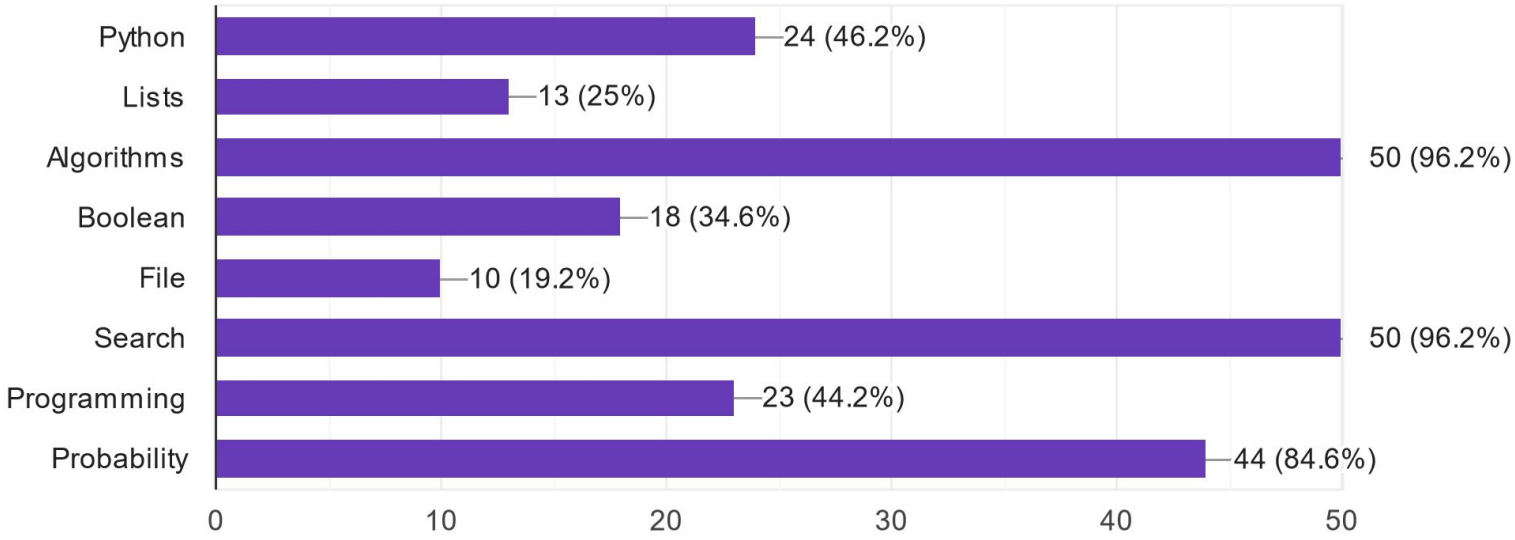
Step 1: Choose features, *e.g.* words

- Python
- List
- Algorithm
- Boolean
- File
- Search
- Programming
- Probability

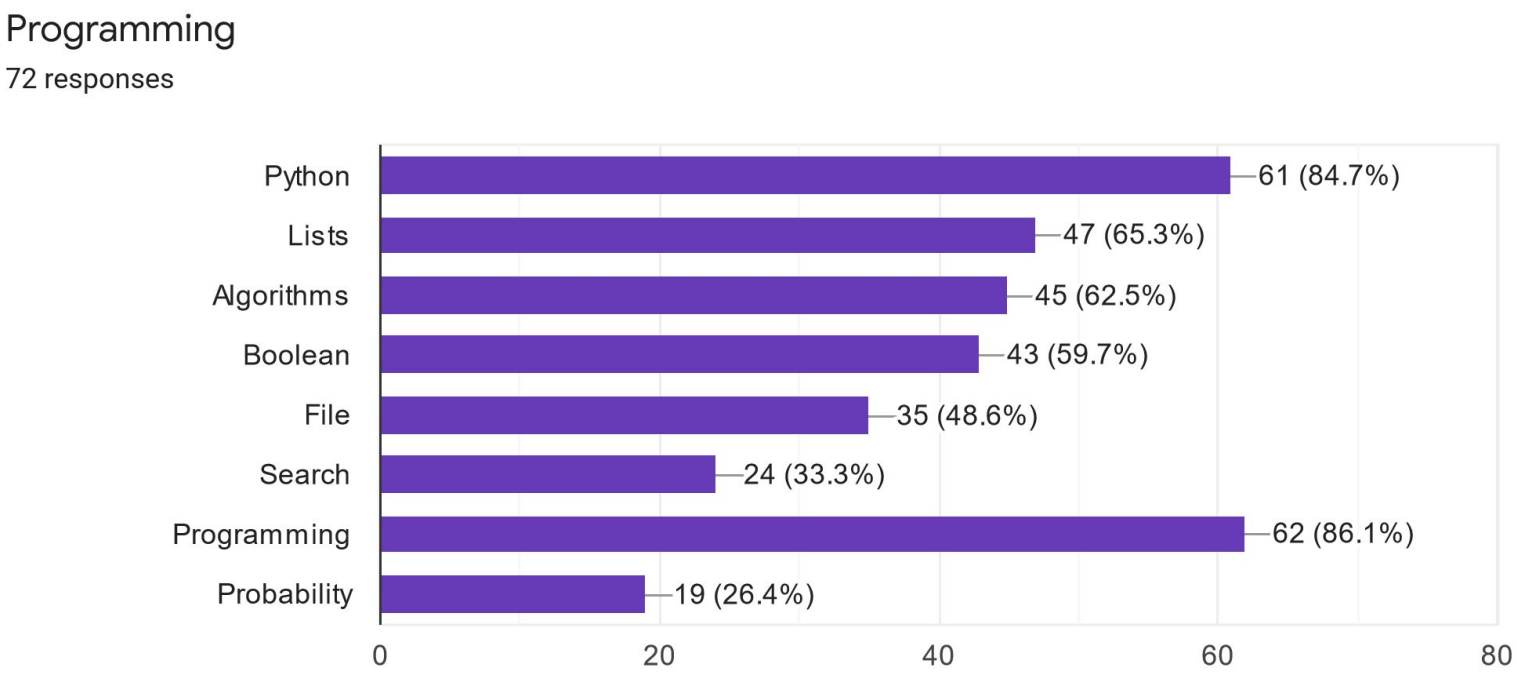
Intelligent Systems Keywords

Choose up to 5 terms that describe the Intelligent Systems course.

52 responses



Programming Keywords



Naive Bayes *Example*

Step 2: Training

- Count occurrences of words
- Split into train/test datasets

Feature	Prog.	Int. Sys
Python	61	24
Programming	62	23
Probability	19	44
Algorithm	45	50
Lists	47	10
Boolean	43	18
File	35	10
Search	24	50
TOTAL	72	52

Naive Bayes *Example*

Step 2: Training

- Class probabilities
- Conditional probabilities
- Smoothing (simple)

Probability	C = IPP	C = IS
P(IS)		42%
P(IPP)	58%	
P(Python|C)	84%	46%
P(Programming|C)	86%	44%
P(Probability|C)	26%	84%
P(Algorithm|C)	62%	96%
P(List|C)	65%	25%
P(Boolean|C)	59%	34%
P(File|C)	48%	19%
P(Search|C)	33%	96%

Naive Bayes *Example*

Step 3: Calculate probability for each class given a new example with features **python, file, lists**.

P(IS|Python, File, Lists)

= P(IS) * P(Python|IS) * P(File|IS) * P(Lists|IS)

= 0.42 * 0.46 * 0.19 * 0.25 = 0.009

Why is the sum not 1?

P(IPP|Python, File, Lists)

= P(IPP) * P(Python|IPP) * P(File|IPP) * P(Lists|IPP)

= 0.58 + 0.84 *0.48 * 0.65 = 0.15

Smoothing

How do we make sure the product of our formula doesn't become zero when the probability of one word is zero?

Smoothing:

- Option A: Add 1 example for each value to your table
- Option B (Laplace): Make the value 0.01 if P(F|X)=0.

If not specified, choose option B.

Probability	C = IPP	C = IS
P(IS)		42%
P(IPP)	58%	
P(Python|C)	84%	46%
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P(Algorithm|C)	62%	96%
P(List|C)	65%	25%
P(Boolean|C)	59%	34%
P(File|C)	48%	19%
P(Search|C)	33%	96%
P(Stefan|C)	0%	100%

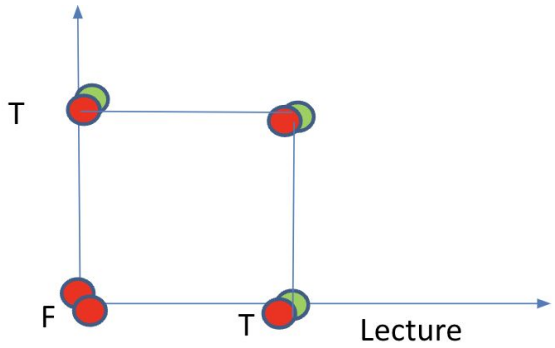
Naive Bayes Classifier *Example Recipe*

1. Preprocessing
   - Choose features, *e.g.* words
   - Split data into training (*e.g.* ⅔) and test (*e.g.* ⅓) sets
2. Training
   - Count occurrences of words
   - Simple Smoothing (+1)
   - Prior probabilities
   - Conditional probabilities
3. Testing
   - Apply Naive Bayes to find class with maximal probability

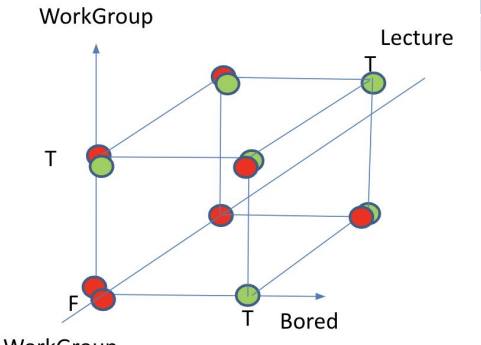
Part 1.2: k-Nearest Neighbour

Joint Probability Table *Revisited*

We can plot this in space…



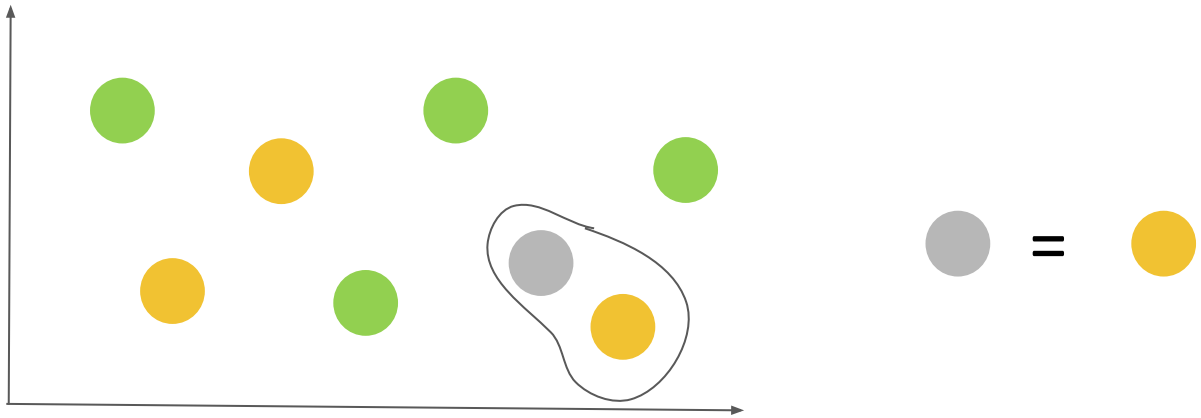
	WorkGroup	Lecture	Pass
John	F	F	F
Peter	F	F	F
Marie	F	T	F
Stefan	F	T	T
Elisa	T	F	F
Mohammed	T	F	T
Rachel	T	T	F
	T	T	T



This works with any continuous variables, *e.g.* age, hours spent.

Nearest Neighbour

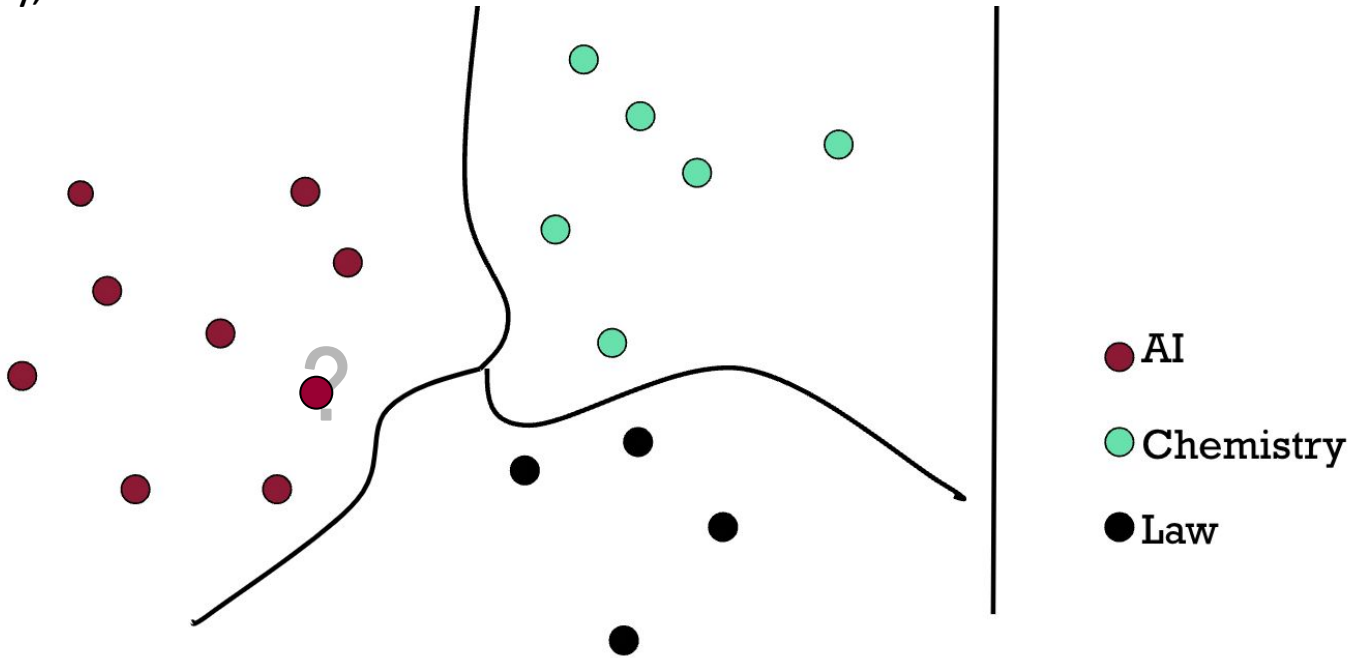
Nearest Neighbour finds the distance between a new example and all the examples in the data which have been labeled already.



…and copies the label from the closest data point.

Nearest Neighbour *Example*

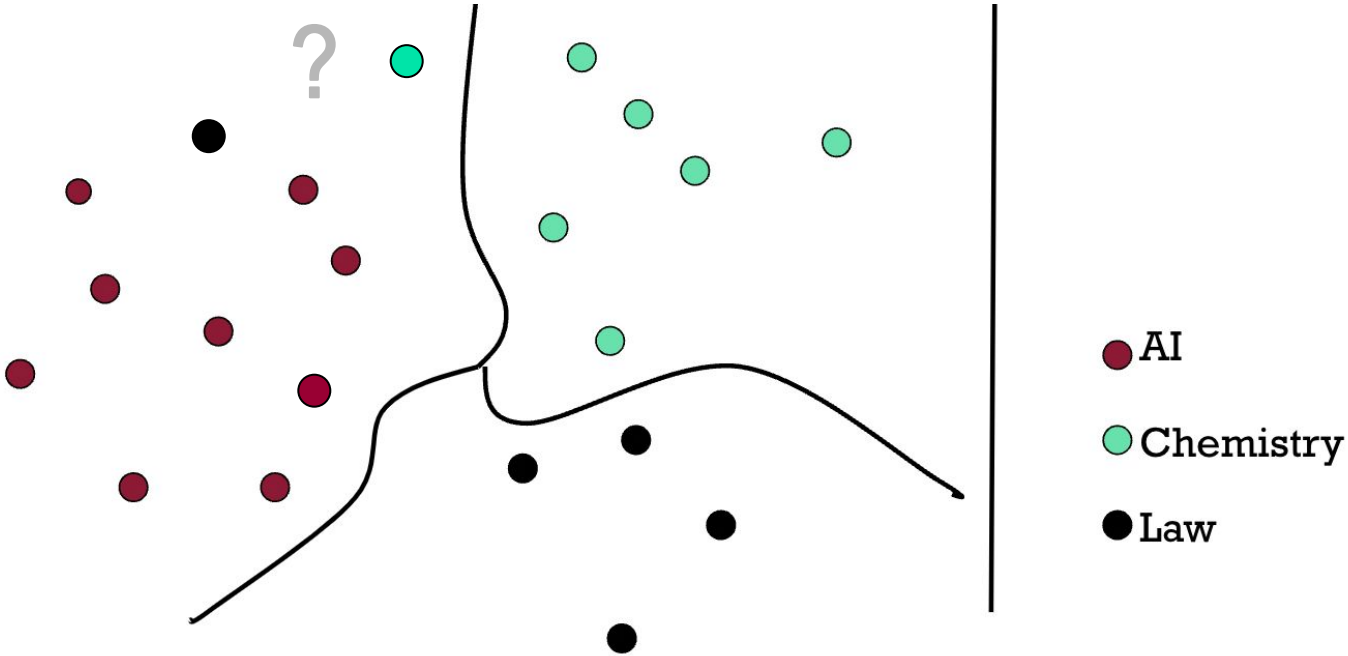
Classify a piece of text according to a set of documents belong to class AI, Chemistry, or Law.



? = “AI”

Nearest Neighbour *Example*

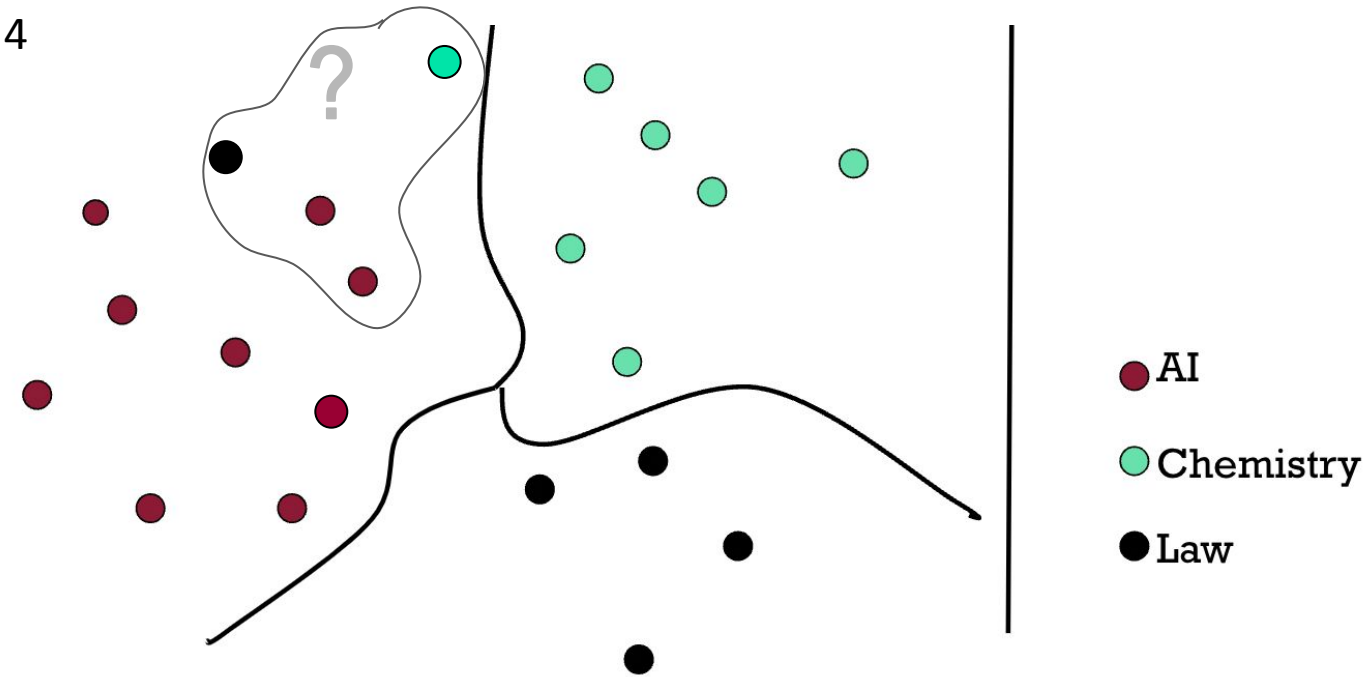
This method can't handle any noise in the data…



Solution: *k*-Nearest Neighbour

Find *k* most similar cases and copy the majority class.

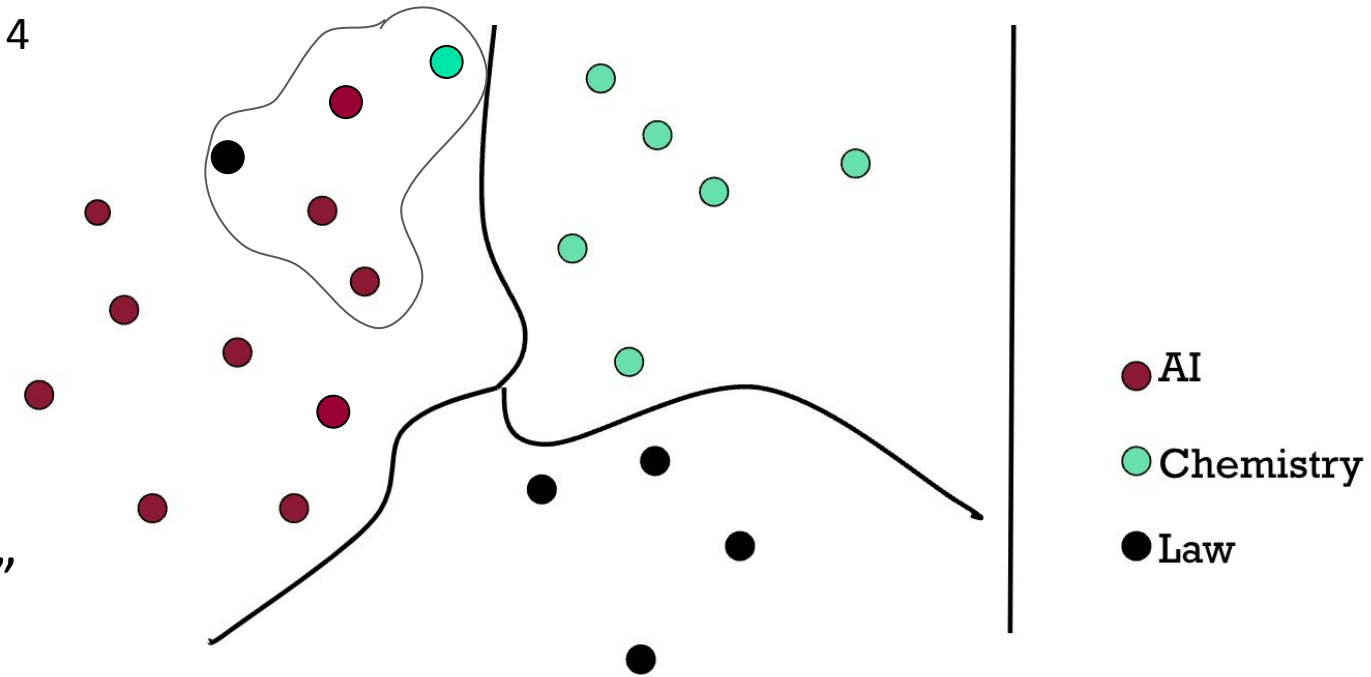
e.g. $k = 4$



Solution: *k*-Nearest Neighbour

Find *k* most similar cases and copy the majority class.

e.g. $k = 4$



? = “AI”

Part 2: Introduction to Machine Learning

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Technology

AI creates gallery of nightmare images for Halloween

25 October 2016 | Technology

Share

The nightmare machine turned pictures from the second presidential debate into darker visions - the images have an impressionistic, sketchy quality because of the way the AI translates them.

The New York Times

ChatGPT's Image Generator | Google's Bard Extensions | How Schools Can Survive A.I. | Smart Ways to Use Chat

THE SHIFT

An A.I.-Generated Picture Won an Art Prize. Artists Aren't Happy.

"I won, and I didn't break any rules," the artwork's creator says.

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Police

Facial recognition could transform policing in same way as DNA, says Met chief

CNN BUSINESS

Audio | Live TV | Log In

Regulators give green light to driverless taxis in San Francisco

By Nadia Bidarian, CNN

Published 6:32 AM EDT, Fri August 11, 2023

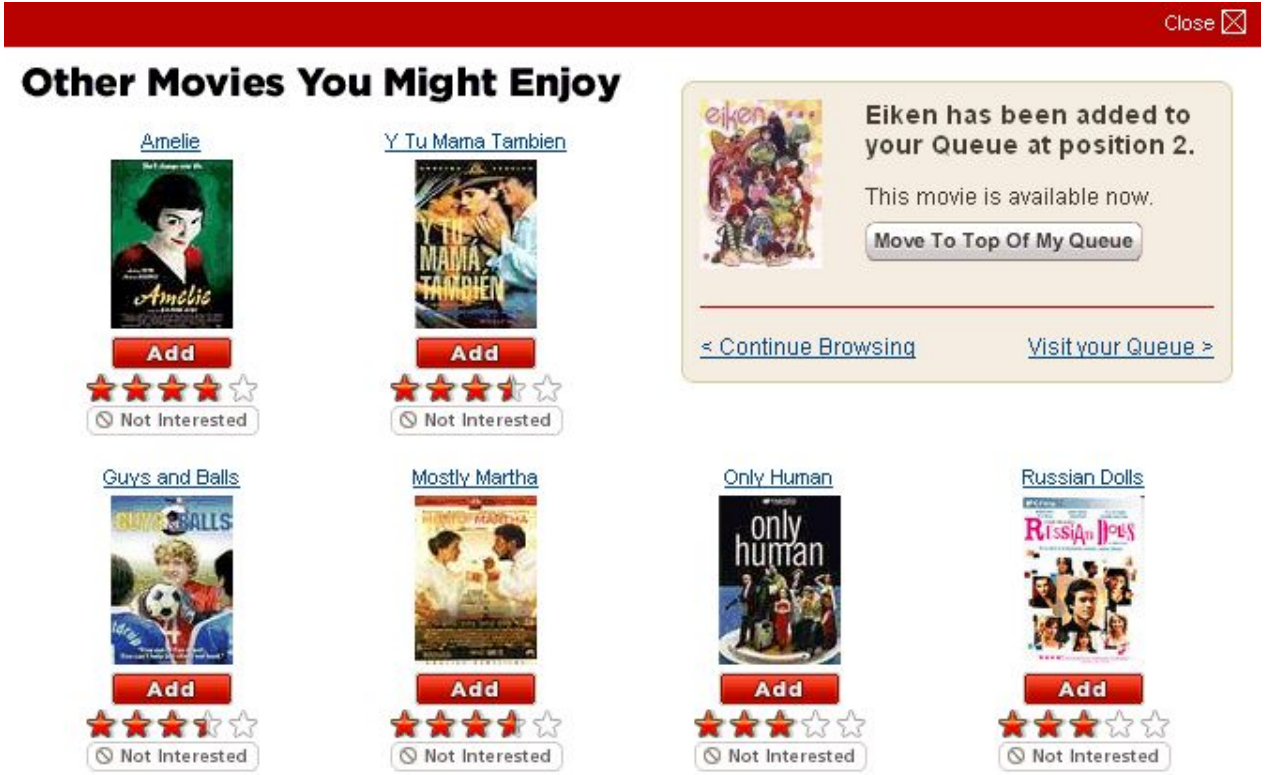
Adaptive Intelligent Agents

Adaptive intelligent agents respond to their environment or other agents and learn from these interactions over time.

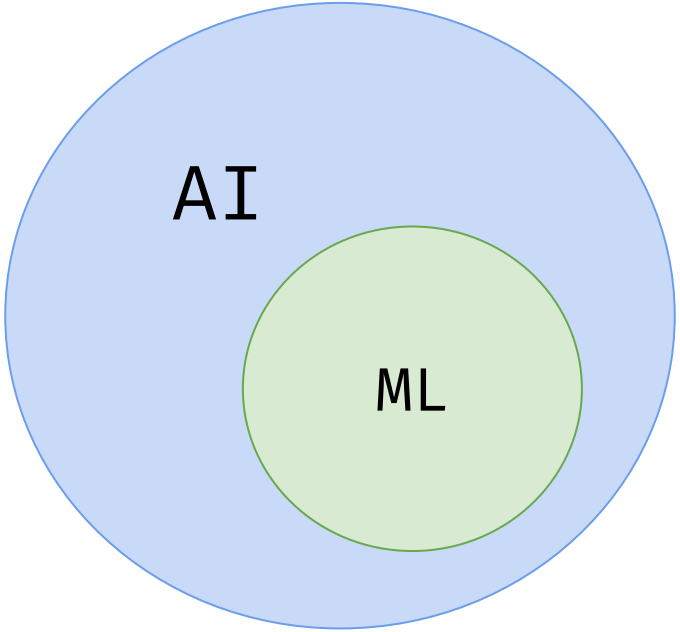
Learning

- Modifies the agent's decision mechanisms
- Is essential for unknown environments
- Useful as a system construction method

Learning Movie Recommendations

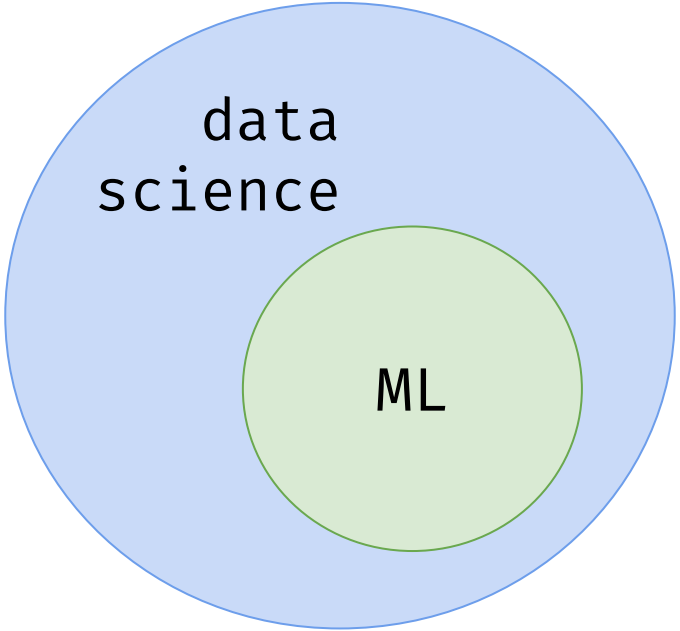


Machine Learning vs. Artificial Intelligence



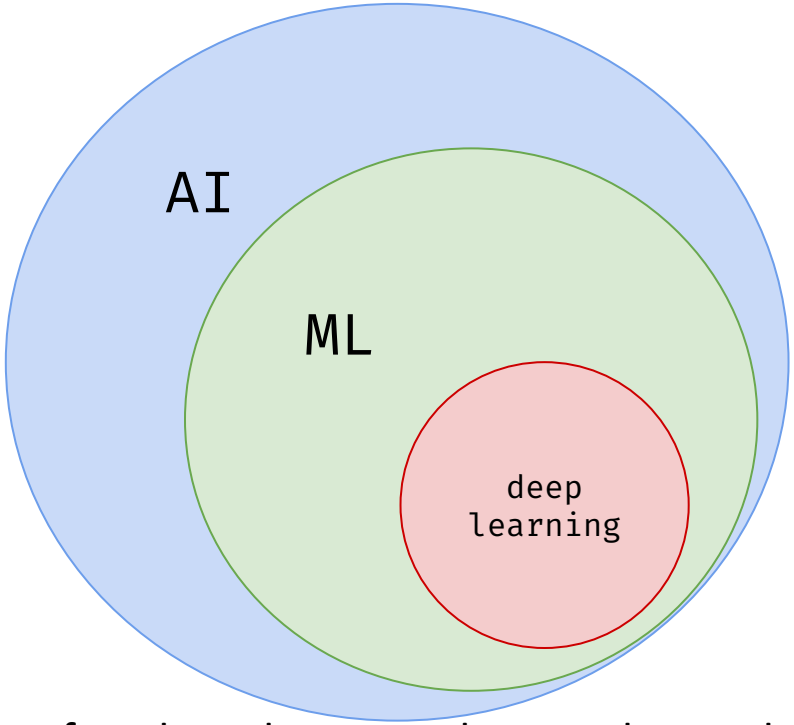
AI but not ML: old chess algorithms, automated reasoning, planning.

Data Science vs. Artificial Intelligence



Data science but not ML: gathering data, harmonising and interpreting data

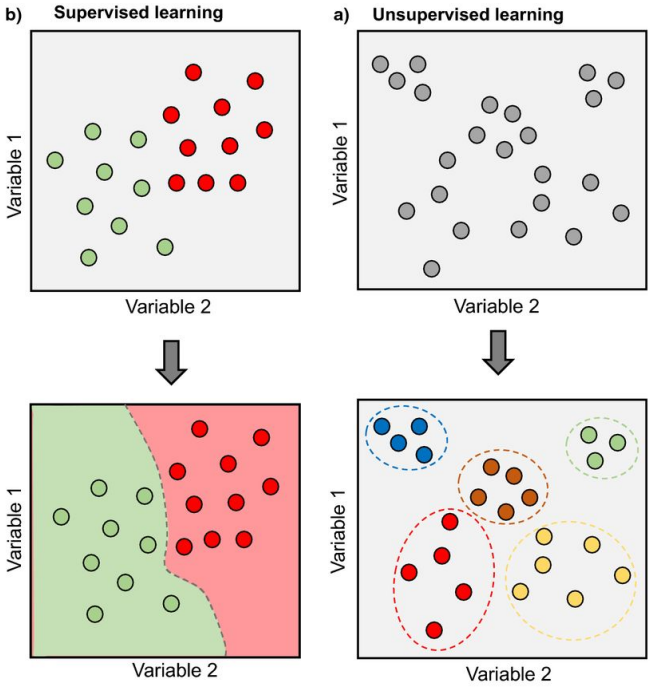
Deep Learning vs. ML vs. AI



Deep learning is a type of ML based on neural networks, modeled similarly to the human brain.

Types of Learning

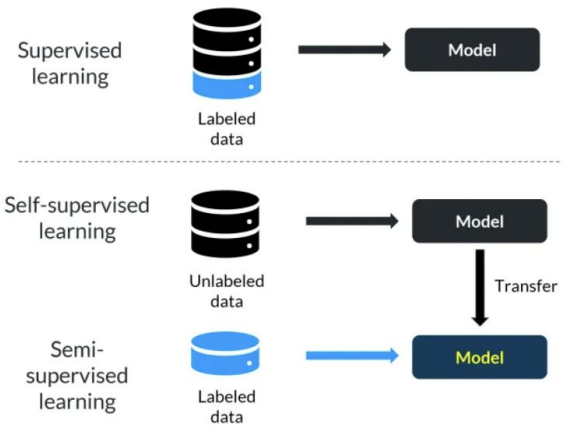
1. Supervised: training data includes correct label for each example
   a. May have only positive examples or positive and negative
2. Unsupervised: training data does not include labels
   a. The model must find the hidden structures in examples by itself



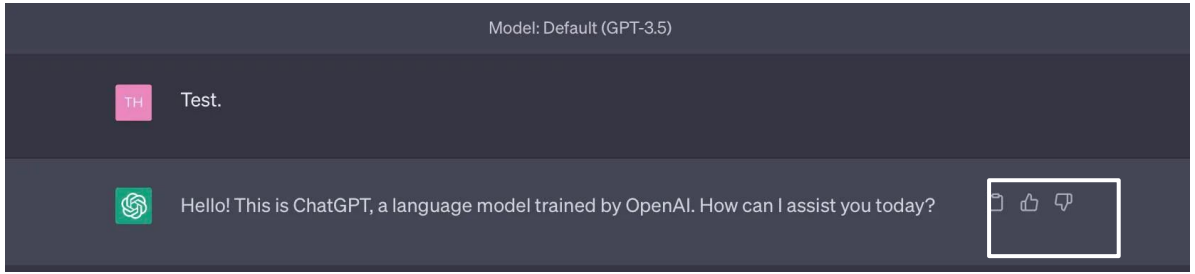
Types of Learning

3. Semi-supervised: partially labeled examples

   a. Active learning: label only high-quality data that improves model performance



4. Reinforcement: desired outputs are rewarded, undesired are punished



Part 2.2: Types of ML Tasks

Types of Learning Problems / Tasks

Within learning types, there are different problems that require different algorithms and give different outputs. Main ML problems include

Supervised	Classification
	Regression
	Ranking
	Collaborative filtering
Unsupervised	Clustering

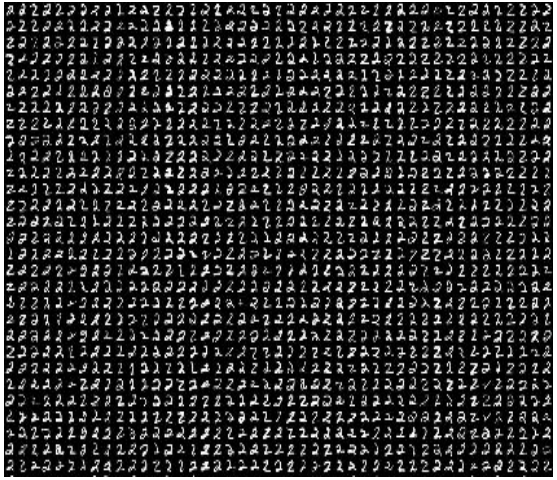
Classification

Type of learning: *supervised*.

Finds the most likely class for each instance.

e.g. handwriting recognition

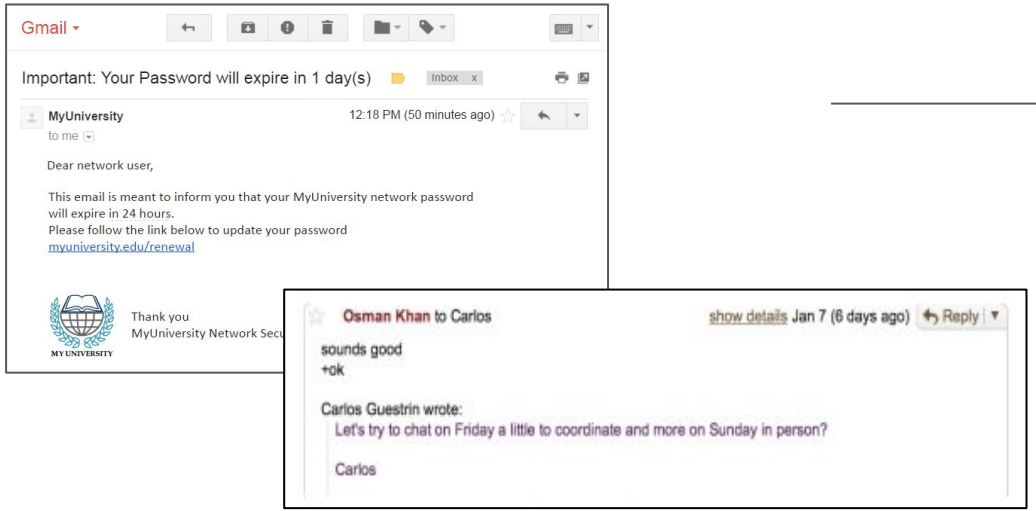
MNIST



Classification *Example*

e.g. spam filtering

DATA



PREDICTION

spam
vs.
not spam

Classification *Example*

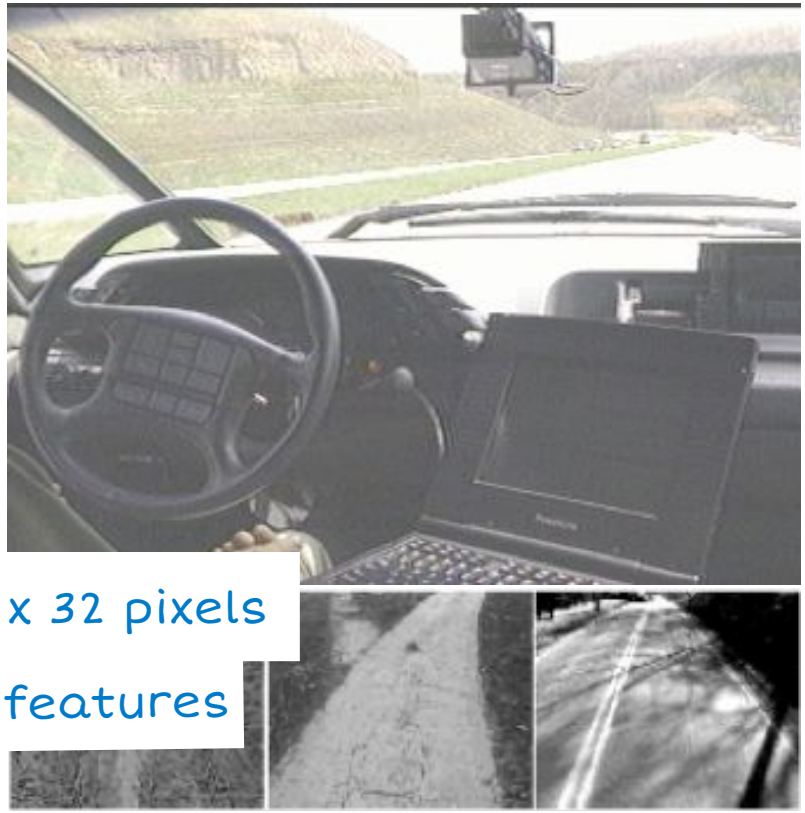
e.g. facial recognition

Example training images
for each orientation

Classification *Example*

e.g. ALVINN (1995)

.3	…	.3	right
.0	…	.0	left
.0	…	.0	left
.1	…	.2	right
.3	…	.4	straight
.1	…	.6	left
.2	…	.3	right
.4	…	.4	left
.6	…	.6	straight
.3	…	.3	straight
.0	…	.0	left

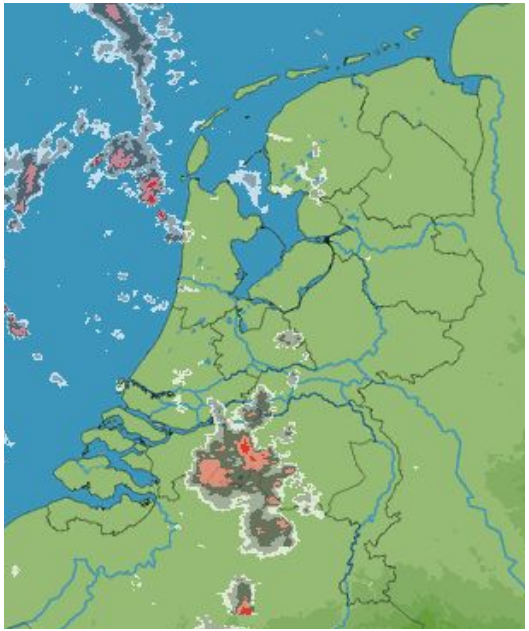


< 30 x 32 pixels

960 features

Classification *Example*

e.g. weather prediction



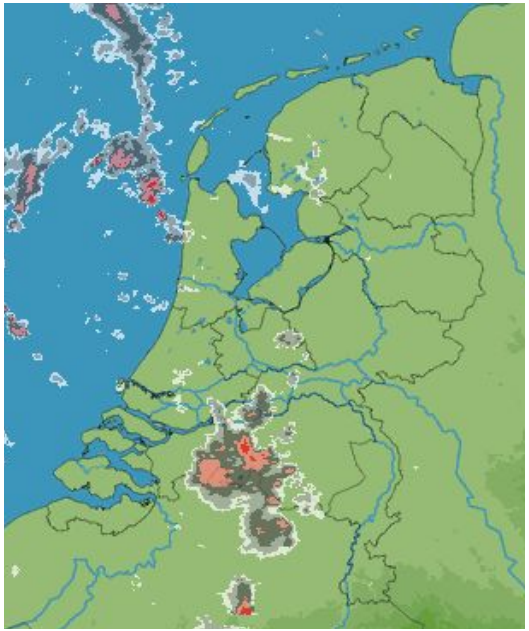
Regression

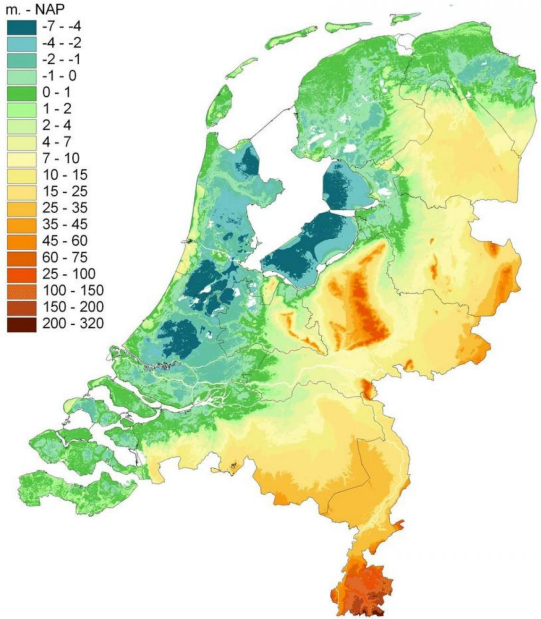
Type of learning: *supervised*. Predict a numeric value for a data point.

e.g. stock market prediction

Regression *Example*

e.g. weather prediction *revisited*

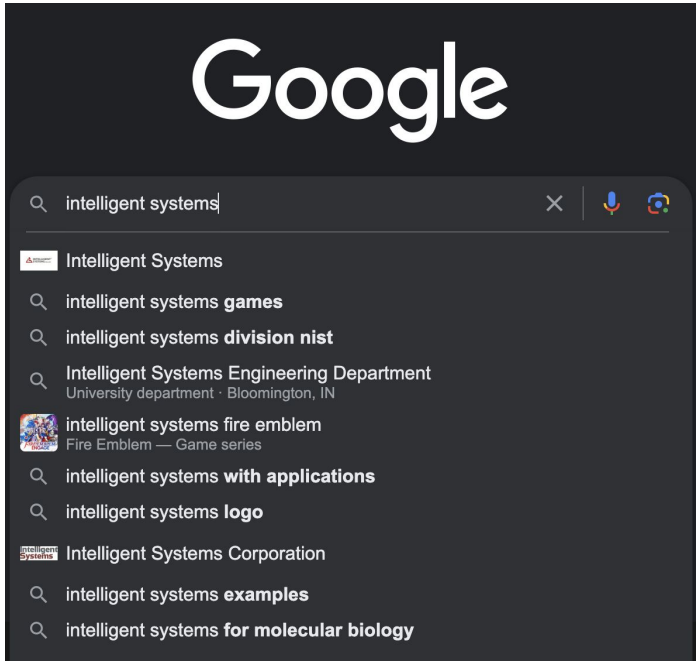




Ranking

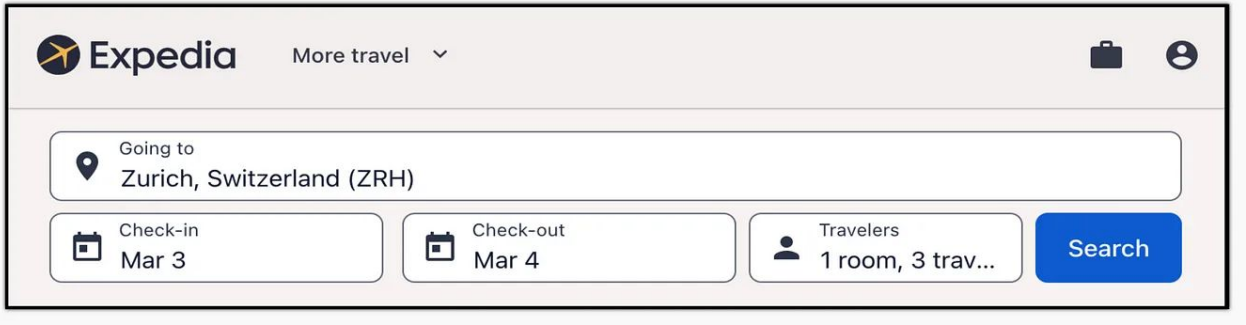
Type of learning: *supervised*. Compare items in order to rank them by a certain criteria.

e.g. web search results

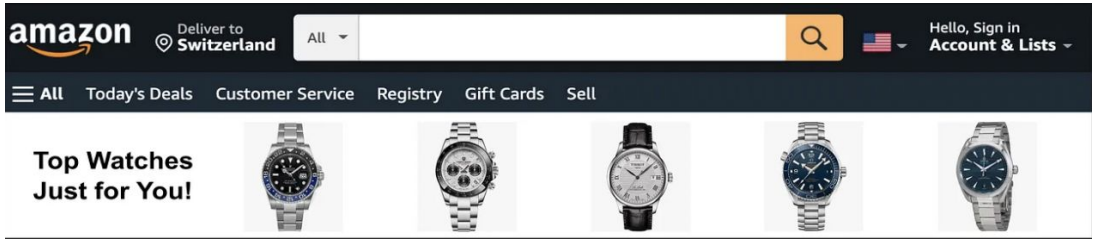


Ranking *Example*

e.g. travel agency results



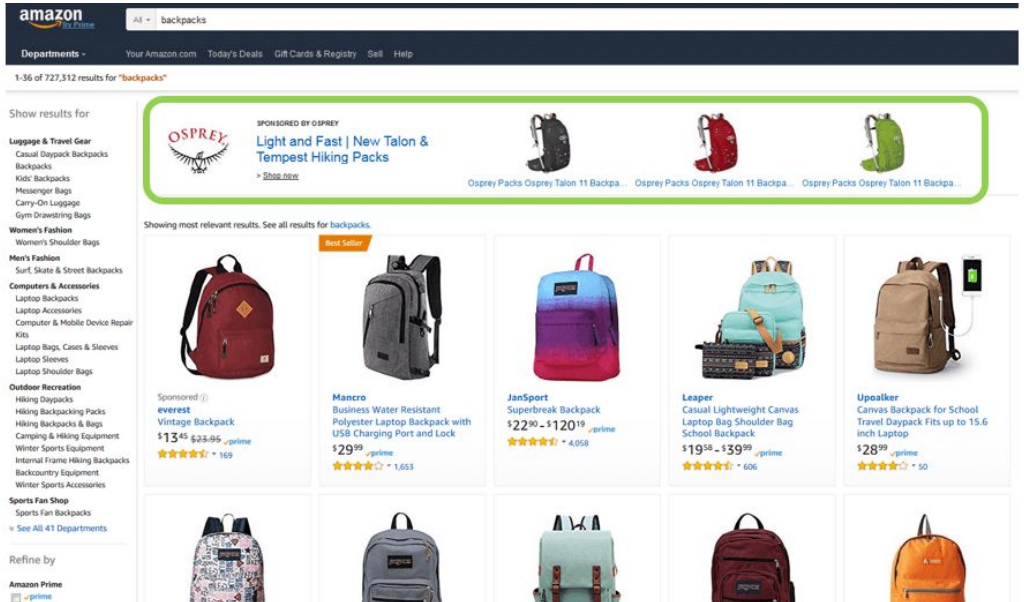
e.g. digital marketplaces



Collaborative Filtering

Type of learning: *(un)supervised*. Filters items that a user might like based on interests of similar users.

e.g. recommender systems



Collaborative Filtering *Example*

Netflix ML competition 2006/09 with $1 million prize.

Leaderboard

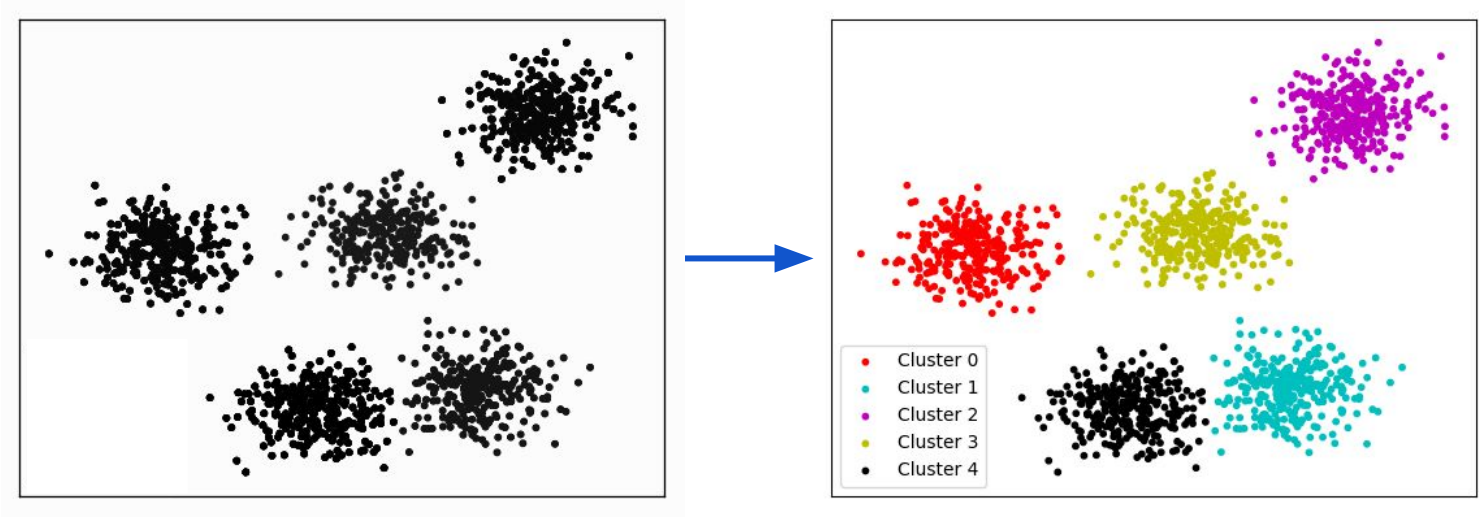
Display top 20 leaders.

Rank	Team Name	Best Score	% Improvement	Last Submit Time
1	The Ensemble	0.8553	10.10	2009-07-26 18:38:22
2	BellKor's Pragmatic Chaos	0.8554	10.09	2009-07-26 18:18:28
Grand Prize - RMSE <= 0.8563				
3	Grand Prize Team	0.8571	9.91	2009-07-24 13:07:49
4	Opera Solutions and Vandelay United	0.8573	9.89	2009-07-25 20:05:52
5	Vandelay Industries !	0.8579	9.83	2009-07-26 02:49:53
6	PragmaticTheory	0.8582	9.80	2009-07-12 15:09:53
7	BellKor in BigChaos	0.8590	9.71	2009-07-26 12:57:25
8	Dace	0.8603	9.58	2009-07-24 17:18:43
9	Opera Solutions	0.8611	9.49	2009-07-26 18:02:08
10	BellKor	0.8612	9.48	2009-07-26 17:19:11
11	BigChaos	0.8613	9.47	2009-06-23 23:06:52
12	Feeds2	0.8613	9.47	2009-07-24 20:06:46
Progress Prize 2008 - RMSE = 0.8616 - Winning Team: BellKor in BigChaos				
13	xiangliang	0.8633	9.26	2009-07-21 02:04:40
14	Gravity	0.8634	9.25	2009-07-26 15:58:34
15	Ces	0.8642	9.17	2009-07-25 17:42:38
16	Invisible Ideas	0.8644	9.14	2009-07-20 03:26:12
17	Just a guy in a garage	0.8650	9.08	2009-07-22 14:10:42
18	Craig Carmichael	0.8656	9.02	2009-07-25 16:00:54
19	J Dennis Su	0.8658	9.00	2009-03-11 09:41:54
20	acmehill	0.8659	8.99	2009-04-16 06:29:35
Progress Prize 2007 - RMSE = 0.8712 - Winning Team: KorBell				
Cinematch score on quiz subset - RMSE = 0.9514				

Clustering

Type of learning: *unsupervised*. Discovering structure in the data to group similar items.

- Requires a criteria for similarity



Clustering *Example*

e.g. image grouping

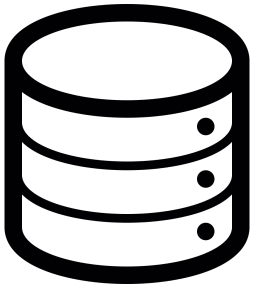
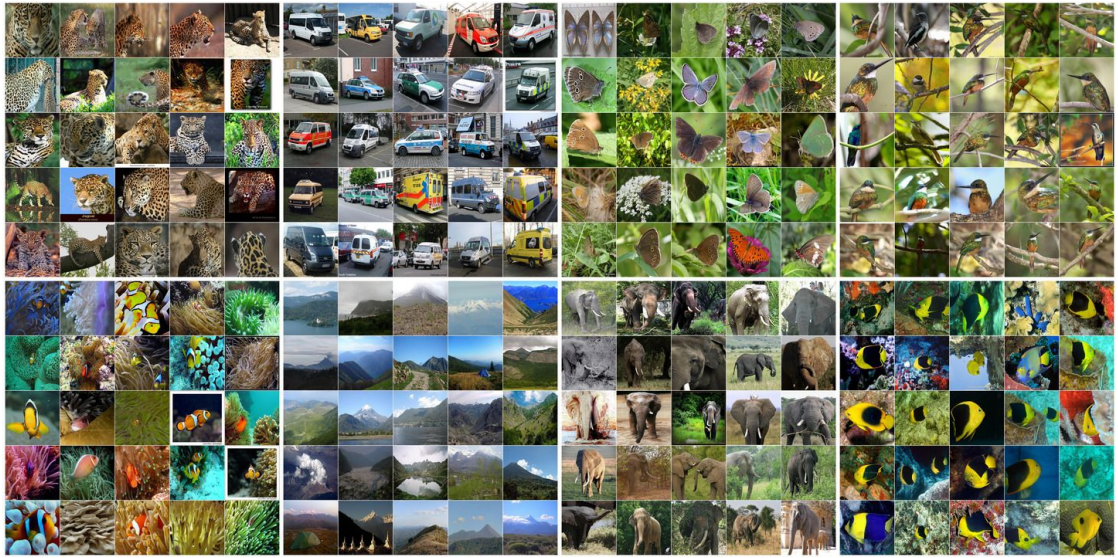


IMAGE DATASET



Clustering *Example*

e.g. grouping similar documents

Intelligent Systems Framework

Efficient, simple, correct, well-understood, generic	Find best decision based on experience.	Weeks 6 & 7
	Find best decision with knowledge & reason.	Weeks 3, 4, 5
	Find best decision in multiple agent setting.	Weeks 1 & 2
	Find best decision with heuristics.	
	Systematically find all possible solutions.	

Summary

Take Home Message

Machine learning allows us to use probabilities to approximate answers and make **predictions**. Different types of **tasks** can deal with different data modalities and outputs.

The four major types of learning specify how we use labels on **training data**, and the five major problem types differ in the predictions they make.

Within classification, we can use **Naive Bayes** to find the most likely class for a data point, or use ***k*-Nearest Neighbour** to find the majority class of a data point's neighbours.